The Right-to-Repair Board Rescuer Deck

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Abstract

Electronic control boards in modern appliances and vehicles are routinely declared unrepairable: a single short-circuited power semiconductor worth a few tens of rupees leads a technician to recommend replacing an entire motherboard costing thousands. This paper presents the Right-to-Repair Board Rescuer Deck, a portable diagnostic platform aimed at neighbourhood repair shops (repair-walas) and DIY consumers. The key insight is that failed power semiconductors expose themselves through localised thermal and electrical signatures when the board is energised at a safe, low voltage. The deck combines a low-cost thermal-imaging array (MLX90640) with automated multimeter probes in a testing tray, and uses edge AI to localise the faulty device, identify its part number, retrieve a datasheet, and display a step-by-step desoldering and replacement guide. The approach reduces repair cost from whole-board replacement to component-level substitution, extending device lifetime and cutting electronic waste.

1. The Average Person's Problem

When a washing machine, a modern microwave, or a car dashboard breaks, the local repairman usually says, "The whole motherboard is dead, you need to buy a new one for ₹15,000." In reality, a single ₹50 power semiconductor (such as a MOSFET or a voltage regulator) has short-circuited. Manufacturers use proprietary software to lock down diagnostics, forcing consumers into expensive replacements.

The gap between the cost of the failed part and the price of the repair is the core economic problem. A transistor or regulator costs between ₹10 and ₹100, yet a board replacement for a washing machine or dashboard can be quoted at ₹8,000 to ₹20,000. Because the board is sealed or undocumented, the technician cannot cheaply prove that the fault is localised, so serviceable hardware becomes electronic waste. Manufacturers reinforce this with proprietary diagnostics, serialised parts, and software locks that reject replacements.

2. The Semiconductor Connection

Modern appliance boards are packed with microcontrollers and power semiconductors. When they fail, they usually fail "short" or "open", creating a detectable thermal or electrical signature.

Power semiconductors fail most often because they carry the largest currents and dissipate the most heat. A MOSFET switching a motor can fail as a drain-to-source short, while a regulator or rectifier can fail short, open, or drift out of specification after mains transients, inrush current, thermal cycling, moisture, or ageing. A shorted device draws abnormal current and becomes a localised hot spot within seconds, whereas an open device removes current from a whole rail, so a safe, current-limited low voltage can reveal either fault.

3. The Proposed Solution

We propose a portable Right-to-Repair Board Rescuer Deck designed for local neighbourhood repair shops (repair-walas) and DIY consumers.

The deck is a portable, laptop-like instrument. Its lid houses a low-cost thermal array (the MLX90640, 32 by 24 pixels) and a camera for part-number recognition; its base holds a testing tray, retractable multimeter probes, and a current-limited power supply. A single-board computer and touchscreen run the diagnosis locally, and the unit is built from widely available parts so that it stays affordable and works offline.

4. How It Works

The deck has a built-in thermal imaging camera (such as an MLX90640 sensor) and automated multimeter probes built into a testing tray. The user places the broken appliance motherboard on the deck, and the deck injects a safe, low voltage into the board. The thermal camera instantly spots the exact semiconductor that is overheating (the shorted chip). The deck's screen then uses edge AI to identify the chip's part number, searches for a datasheet, and shows a 3D video guide on how to desolder and replace that exact ₹50 chip.

In use, the operator places the suspect board on the tray, where spring-loaded probes contact its main power rails, and the deck energises it at a safe, current-limited voltage while capturing a thermal image. A Python and OpenCV pipeline compares the frame with the ambient baseline, flags regions that exceed a temperature threshold, and overlays their coordinates. The hottest anomaly is almost always the shorted device, because that is where electrical energy becomes heat. The deck then reads the part number, retrieves its datasheet, and shows a three-dimensional guide for replacing it.